# Romit Chakraborty

+1 773-816-8338 | romit@pointreyessound.com | www.pointreyessound.com

## Professional Profile

Quantum chemist and architect of phase-space electronic structure theory, redefining the chemical bond as a dynamic probability fluid. Uniting quantum mechanics with kinetic transport, my research pioneers Quantum Boltzmann Equation (QBE) solvers that autonomously regularize mean-field singularities and ameliorate the exponential scaling walls of strongly correlated quantum matter. Building on postdoctoral research at UC Berkeley and doctoral training at the University of Chicago, I now direct Point Reyes Sound, Inc., translating thermodynamic theory into scalable, hardware-agnostic architectures for next-generation quantum simulation.

## Professional Experience

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| June 2026 | **Founder & CEO**, Point Reyes Sound, Inc., San Francisco, CA |
| 2024-2025 | **Senior Computational Scientist, AI & Solutions**, PsiQuantum, Palo Alto, CA, USA |
| 2023-2024 | **Staff Researcher in Quantum AI**, University of Chicago, Chicago, IL, USA |
| 2017-2023 | **Postdoctoral Research Fellow (Computational Science)**, UC Berkeley & LBNL, Berkeley, CA, USA |
| 2018-2020 | **Scientific Software Developer**, Q-Chem Inc., CA, USA |

## Education

**University of California, Berkeley and Lawrence Berkeley National Laboratory** — *Berkeley, CA, USA*
**Postdoctoral Research Fellow in Computational Materials Science** — *2017-2023*
Advisor: Prof. Martin Head-Gordon

**University of Chicago** — *Chicago, IL, USA*
**MS, Ph.D. in Theoretical and Computational Quantum Chemistry** — *2012-2017*
Advisor: Prof. David A. Mazziotti

**IIT Bombay** — *Bombay, India*
Master of Science — *2009-2012*
Thesis: Relativistic Quantum Mechanics of the Pionium Atom in Uniform Magnetic Fields

**St Stephen's College, Delhi** — *Delhi, India*
B.Sc (Hons) in Chemistry, with Physics and Mathematics — *2006-2009*

## Academic & Research Appointments

**Point Reyes Sound, Inc.** — *San Francisco, CA*
**Founder & CEO** — *June 2026 -- Present*

- **Phase-Space Electronic Structure:** Formulated a Hessian-free orbital optimization framework utilizing continuous phase-space fluid advection and the Bhatnagar-Gross-Krook (BGK) collision operator to autonomously regularize mean-field singularities.
- **Quantum Boltzmann Entrainment Tensor (QBET):** Developed a theoretical methodology to restrict quantum simulation workloads strictly to the active space, effectively resolving exponential scaling limits in strongly correlated multireference systems.
- **Solid-State $k$-Space Decoupling:** Generalized the phase-space transport engine for periodic crystalline systems by projecting continuous Wigner distributions onto cGTOs, mapping multidimensional transport to decoupled matrix relaxations across the Brillouin zone.

**PsiQuantum, Corp.** — *Palo Alto, CA*
**Senior Computational Scientist** — *Oct 2024 -- Mar 2025*

- Translated quantum chemical models of correlated electron systems into robust algorithmic frameworks with scalable software architectures.

**University of Chicago** *Chicago, IL*

**Staff Researcher** *Nov 2023 -- Oct 2024*

- **Physics-Grounded Generative Workflows:** Led the development of a symmetry-aware framework conditioning generative models on strict physical constraints (molecular geometry, charge/spin manifolds, point-group symmetry) to resolve out-of-distribution mode collapse.
- **Physical Symmetry Integration:** Pioneered the integration of physical symmetries into large-scale generative architectures, securing Microsoft Azure Foundation Models Research funding to bridge theoretical chemistry and machine learning.
- **Mathematical Verification Metrics:** Engineered rigorous protocols for generating mathematically-verified synthetic data, establishing evaluation metrics defined by exact structural equivalences rather than heuristic approximations.

# Publications

$^{+}$ *Equal Contributors*

## Current Research and Technical Reports

**Chakraborty, R. Rendering the Quantum Fluid: Mapping Phase-Space Electronic Structure to Tile-Based Deferred Shading**, *Point Reyes Sound Technical Whitepaper PRS-TR-2026-02,* ***2026***. [PDF] [Interactive Dossier]

**Chakraborty, R. The Quantum Boltzmann Entrainment Tensor: A Hessian-Free Framework for Multireference Orbital Optimization**, *Point Reyes Sound Technical Report PRS-TR-2026-01 / In Preparation for JCTC,* ***2026***. [PDF] [Interactive Dossier]

**Chakraborty, R. Quantum Resource Estimation: Hardware Resource Compilation for Active-Space Driven Fault-Tolerant Quantum Simulation**, *Point Reyes Sound Technical Whitepaper PRS-TR-2026-03,* ***2026***. [PDF]

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## Peer Reviewed Articles and Preprints

**Chakraborty, R.** ***2026***. **Quantum Boltzmann Equation Self-Consistent-Field for the Entropic Regularization of Mean-Field Singularities**. *arXiv:2608.14979 [quant-ph, physics.chem-ph].* [arXiv:2608.14979] [PDF]

**Chakraborty, R.**, Talbot, J. J., Shen, H., Yabuuchi, Y., Jiang, H., Carsch, K.M., Furukawa, H., Long, J.R., Head-Gordon, M. ***2024***. Quantum Chemical Modeling of Hydrogen Binding in Metal-Organic Frameworks: Validation, Insight, Predictions and Challenges. *Selected as a Hot Article*. Phys. Chem. Chem. Phys. 26 (8), 6490-6511

Carsch, K.M.,**Chakraborty, R.**, Head-Gordon, M, Long, J.R. et. al. . ***2023***. Selective Adsorption of Oxygen from Humid Air in a Metal–Organic Framework with Trigonal Pyramidal Copper(I) Sites. J. Am. Chem. Soc. 2024, 146, 5, 3160–3170

Talbot J.J., **Chakraborty, R.**, Shen, H., Head-Gordon, M. ***2023***. A Free Energy Decomposition Analysis: Insight into Binding Thermodynamics from Absolutely Localized Molecular Orbitals. *J. Phys. Chem. Lett.*, 14, 5432-5440.

Funke L.M., **Chakraborty R.**, Carsch, K.M., Head-Gordon, M., Long, J.R., Reimer, J.A..***2023***. Assessment of Adsorbate $\pi$-backbonding in Cu(I) Metal–Organic Frameworks via Multinuclear NMR Spectroscopy and Density Functional Theory Calculations. *J. Phys. Chem. C* 127 (15): 7513–7519

**Chakraborty, R.**, Carsch, K.M., Jaramillo, David E, Yabuuchi Tuto, Furukawa H, Long, J.R., Head-Gordon M. ***2022***. Prediction of Multiple Hydrogen Ligation at a Vanadium(II) Site in a Metal-Organic Framework. *J. Phys. Chem. Lett.*, 13(44): 10471-10478.

Barnett, B.R., Evans, H.A, Su, G.M., Jiang, H.Z.H., **Chakraborty, R.**, Banyeretse, D., Hartman, T.J., Martinez, M.B., Trump, B.A., Tarver, J.D., Dods, M.N., Funke, L.M., Borgel, J., Reimer, J.A., Drisdell, W.A., Hurst, K.E., Gennett, T., FitzGerald, S.A., Brown, C.M., Head-Gordon, M., Long, J.R. ***2021***. Observation of an Intermediate to H2 Binding in a Metal–Organic Framework. 2021. *J. Am. Chem. Soc.*, 143(36): 14884-14894.

Epifanovsky E., Gilbert A.T.B., Feng X., Lee J., Mao Y., Mardirossian N., ..., **Chakraborty, R.**, ..., Head-Gordon, M., Herbert J.M., Krylov, A.I. Software for the frontiers of quantum chemistry: An overview of developments in the Q-Chem 5 package. ***2021***. *J. Chem. Phys.*, 155(8):084801

Jaramillo, D.E., Jiang, H.Z.H., Evans, H.A., **Chakraborty, R.**, Furukawa, H., Brown, C.M., Head-Gordon, M., Long, J.R. ***2021***. Ambient-temperature hydrogen storage via vanadium (II)-dihydrogen complexation in a metal–organic framework. *J. Am. Chem. Soc.*, 143(16):6248:6256

Stauch T.[+], **Chakraborty, R.**[+], Head-Gordon, M. 2019. Quantum Chemical Modeling of Pressure Induced Spin Crossover in Octahedral Metal Ligand Complexes. ***2019***. *ChemPhysChem*, 20(21):2742-2747

**Chakraborty, R.**, Mazziotti, D.A. ***2018***. Sparsity of the wavefunction from the generalized Pauli exclusion principle. *J. Chem. Phys.* 148 (5), 054106.

**Chakraborty, R.**, Mazziotti, D.A. 2017. Noise-assisted energy transfer from the dilation of the set of one electron reduced density matrices. ***2017***. *J. Chem. Phys.* 146 (18), 184101.

**Chakraborty, R.**, Mazziotti, D.A. ***2016***. Role of the generalized Pauli constraints in the quantum chemistry of excited states. *Int. J. Quantum Chem.* 116 (10), 784-790.

**Chakraborty, R.**, Mazziotti, D.A. ***2015***. Structure of the one electron reduced density matrix from the generalized Pauli exclusion principle. *Int. J. Quantum Chem.* 115 (19), 1305-1310.

**Chakraborty, R.**, Mazziotti, D.A. ***2015***. Sufficient condition for the openness of a many electron quantum system from the violation of a generalized Pauli exclusion principle. *Physical Review A* 91 (1), 010101

Datta, S. N., Ghosh A., **Chakraborty, R.**. ***2015***. Variational aspects of the Klein–Gordon equation. *Ind. J. Phys.* 89 (2), 181-187.

**Chakraborty, R.**, Mazziotti, D.A. ***2014***. Generalized Pauli conditions on the spectra of one electron reduced density matrices of atoms and molecules. *Phy. Rev. A* 89 (4), 042505.

`https://scholar.google.com/citations?hl=en&user=m4HlFRIAAAAJ&view_op=list_works&sortby=pubdate`

## Awards, Fellowships, & Grants

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| Fall, 2023 | **Accelerate Foundation Models Research**, Microsoft Research for LoRA fine-tuning OpenAI models |
| Smr, 2023 | **Award for Contributions to Inclusion in Science and Society**, Berkeley Lab (Postdoc Assoc.) |
| Smr, 2023 | **Successful EOI for Independent Early Career Fellowships**, University of Oxford, UK |
| Fall, 2015 | **Arts Science and Culture Graduate Fellowship**, University of Chicago, USA |
| Smr, 2009 | **Lindau Graduate Fellowship**, German Research Foundation (DFG) |

## Outreach & Professional Development

### Peer Review

***Nature Communications, 2022***
***Journal of Chemical Theory and Computation, 2023***
***Molecular Physics, 2018, 2019, 2020***
***Journal of Physics: Condensed Matter, 2017***

### Professional Memberships

American Physical Society
Councillor (alternate) at California Section of the American Chemical Society (2023-2026) Program Book Coordinator at the Western Regional Meeting (2025)
Project SEED (Summer Opportunities for Economically Disadvantaged Students) (2025-2026)
Member at Large, Cal ACS 2026-2027